

Python merge-service fixes “ search-flow audit 2026-06-15

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Anti-bluff (Â§11.4): every fix RED-first (test reproduces the defect on the pre-fix code), permanent Â§11.4.135 regression guard added, GREEN proven by pytest + LIVE curl against the running stack.

BUG-1 “ provider selection (CRITICAL)

Root cause: only “search all enabled providers” existed; no wire field, no orchestrator param, no plugin filter for single/subset. Single/selected UI modes silently behaved as “all”.

Fix: - `api/routes.py` “ SearchRequest gains trackers: `list[str] | None = None`; POST `/search` and POST `/search/sync` pass `request.trackers` down. - `merge_service/search.py` “ new helper `_select_trackers(self, tracker_filter)` (calls `_get_enabled_trackers()`, then if filter non-empty keeps only trackers whose `.name` is in the case-insensitive filter set). `start_search(... trackers=None)` persists the filter on `SearchMetadata.tracker_filter`; `_run_search` reads it back via `_select_trackers`; `search(... trackers=None)` threads it through. `Empty/None == all` (back-compat).

RED tests: `tests/unit/merge_service/test_provider_selection.py` - `test_single_provider_filters_fanout` “ trackers=[“ruTracker”] “ dispatched set == {“ruTracker”}, `metadata.trackers_searched` == [“ruTracker”]. - `test_subset_provider_filters_fanout` “ two-element subset. - `test_none_filter_fans_out_to_all` / `test_empty_filter_fans_out_to_all` “ back-compat. - `test_filter_is_case_insensitive` “ “RuTracker” “ {“ruTracker”}. - `test_unknown_name_yields_empty_fanout_gracefully` “ unknown name “ zero dispatch, no crash. - `test_start_search_seeds_filtered_trackers` “ async POST path seeds only the subset. Asserts on actual dispatched tracker set + `metadata.trackers_searched`, not status codes.

BUG-6 “ integer-size crash in dedup fallback

Root cause: `deduplicator._parse_size_to_bytes` passed a raw int into `re.match` (no `str()`), raising `TypeError`; `unguarded_update_best_quality` “merge_results aborted the whole merge “ zero results.

Fix: `merge_service/deduplicator.py` `_parse_size_to_bytes` coerces `str(size_str)` after the falsy guard (mirrors the `api.routes/_parse_size` fix). `_fallback_quality` signature widened to object.

RED tests: `tests/unit/merge_service/test_dedup_integer_size.py` - `test_parse_size_to_bytes_accepts_integer` / `_accepts_negative_sentinel`. - `test_fallback_quality_survives_integer_size`. -

test_merge_survives_integer_size” forces the api.routes ImportError branch, int+(-1) sizes; asserts non-empty merge.

BUG-7 “all-tracker failure indistinguishable from empty

Root cause: every-provider-fail completed as status=“completed”, total_results=0 “ dashboard “No results found.”, indistinguishable from genuinely empty.

Fix: merge_service/search.py _run_search “ when merged==0 AND total_results==0 AND every searched tracker stat is error/timeout, set status = “all_trackers_errored” (else “completed”). Per-tracker errors already accumulate in metadata.errors.

RED tests: tests/unit/merge_service/test_all_trackers_error.py - test_all_trackers_error_is_distinguishable “ all raise “ status != “completed”, len(errors)==3. - test_genuinely_empty_stays_completed “ no errors + 0 results stays “completed”, errors==[].

VERIFY

pytest GREEN (the 3 new regression files)

```
$ .venv/bin/python -m pytest tests/unit/merge_service/test_provider_select
    tests/unit/merge_service/test_dedup_integer_size.py \
    tests/unit/merge_service/test_all_trackers_error.py -v --import-mode=i
...
===== 13 passed in 0.78s =====
```

RED proof (pre-fix, same files): 12 failed / 1 passed “ provider tests reject the trackers= kwarg, BUG-6 raises TypeError: expected string or bytes-like object, got 'int', BUG-7 reports status==“completed”.

Full merge_service unit suite “ no regression

```
$ .venv/bin/python -m pytest tests/unit/merge_service/ -q --import-mode=im
818 passed, 3 warnings in 84.99s
```

ruff: All checks passed! on the 3 source + 3 test files. (Pre-existing, unrelated: tests/unit/test_plugin_yts_deep.py fails 45 on a pristine stashed tree “ collection error in the YTS plugin test, not touched by these fixes.)

LIVE “ running stack (podman VM, merge service :7187)

After editing: cleared __pycache__, podman restart qbittorrent-proxy, verified served code matches committed edits (_select_trackers, all_trackers_errored, size_str = str(size_str) all present in /config/download-proxy/“).

```
=== FILTERED trackers_searched (rutor only) ===
status: completed | total: 3 | trackers_searched: ['rutor']
```

```
=== NO-FILTER (all enabled providers) ===
status: completed | total: 1182 | n_trackers: 29 | trackers_searched:
```

```
['rutracker','kinozal','nnmclub','iptorrents','jackett','academictorrents','anilibra','bitsearch','gamestorrents','glotorrents','kickass','limetorrent','linuxtracker','megapeer','nyaa','piratebay','pirateiro','rockbox','rutor','snowfl','tokyotoshokan','torlock','torrentdownload','torrentgalaxy','torrentkitty','torrentproject','torrentscsv','yourbittorrent','yts']
```

Filtered search hits ONLY rutor (3 results); no-filter hits all 29 providers (1182 results) – provider selection proven end-to-end through the real HTTP path.

Files changed: download-proxy/src/api/routes.py, download-proxy/src/merge_service/search.py, download-proxy/src/merge_service/deduplicator.py. Tests added: tests/unit/merge_service/test_provider_selection.py, test_dedup_integer_size.py, test_all_trackers_error.py. NOT committed.